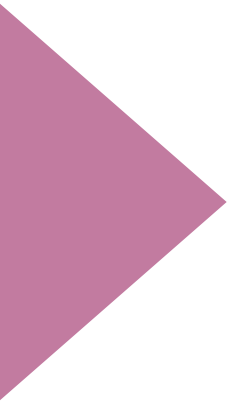


Amazon VPC-4



Table of Content

- WORDPRESS WITH LAMP STACK ON VPC
- NACL TABLES



WORDPRESS WITH LAMP STACK ON VPC

Dynamic Website

Dynamic Website



Operating
System

Web Server

Database

Prg. Language

Setup Wordpress with Database

LAMP:



Linux



Apache



MySQL



PHP

Operating System

Web Server



Database

Progr. language

User Data



LAMP:

Installed-ready



EC2 Amazon Linux 2

User Data

User Data

User Data

User Data





10.7.0.0/16



Internet Gateway

VPC

us-east-1a

AZ

us-east-1a-Public

NAT
Instance

NAT
Gateway



Route Table

EC2

No public IP

us-east-1a-Private

us-east-1b

AZ

us-east-1b-Public

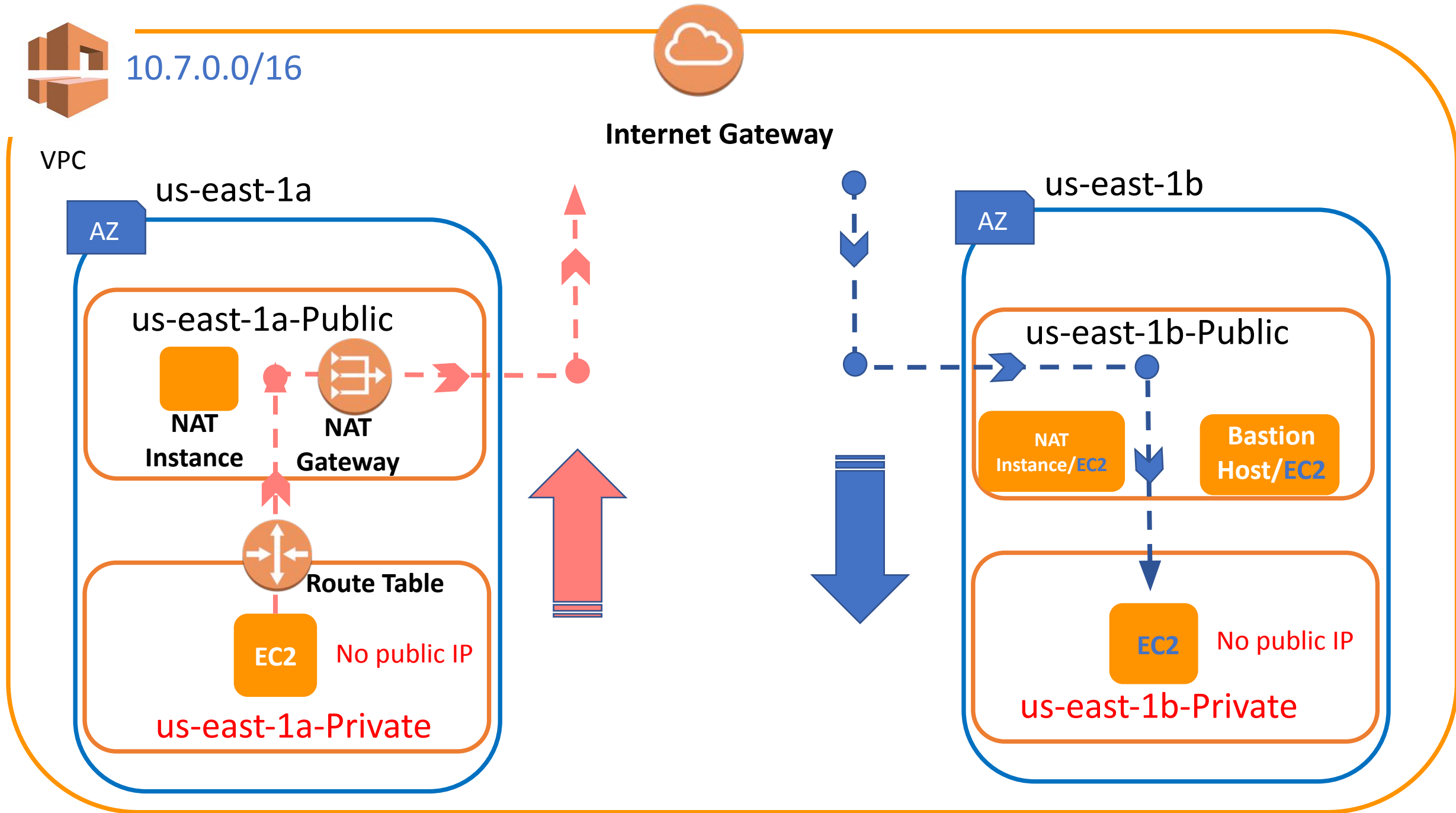
NAT
Instance/EC2

Bastion
Host/EC2

EC2

No public IP

us-east-1b-Private



Operating
System

Web Server



Database

Progr. language

User Data

LAMP:



Installed-ready



EC2 Amazon Linux 2

User Data



?

User Data



User Data



It is in another instance in the Private Subnet



Cloud



Region



Clarus-VPC-a

Availability Zone 1-a

Public Subnet 1a

Private Subnet 1a

1- Desired Scenario



Internet Gateway

Availability Zone 1-b

Public Subnet 1b



Private Subnet 1b

Availability Zone 1-c

Public Subnet 1c

Private Subnet 1c



Cloud

Region



Clarus-VPC-a

1- Desired Scenario



Internet Gateway

Availability Zone 1-a

Availability Zone 1-b

Availability Zone 1-c

Public Subnet 1a



Private Subnet 1a

Public Subnet 1b



Private Subnet 1b

Public Subnet 1c



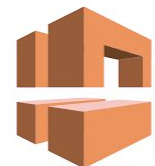
Private Subnet 1c



Cloud



Region

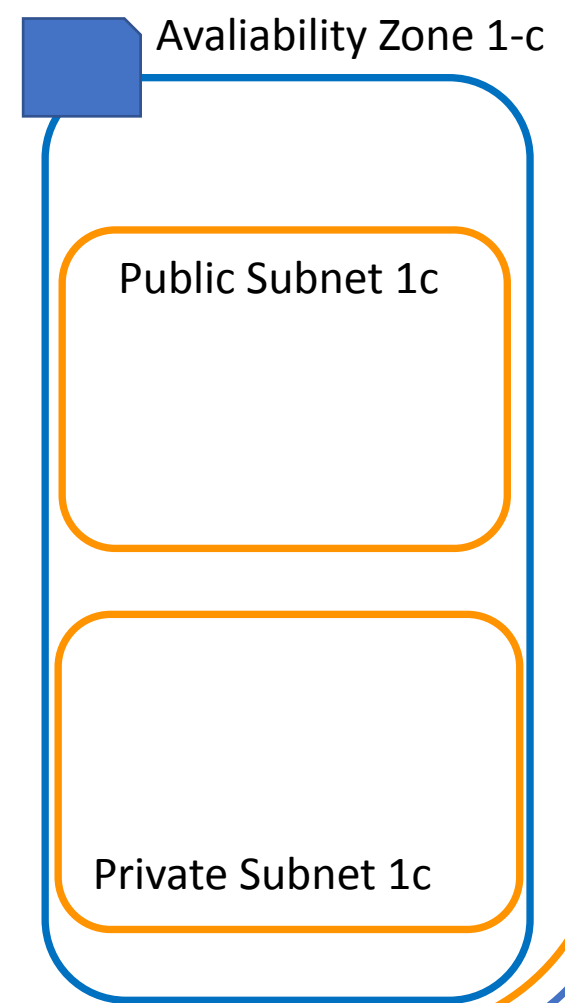
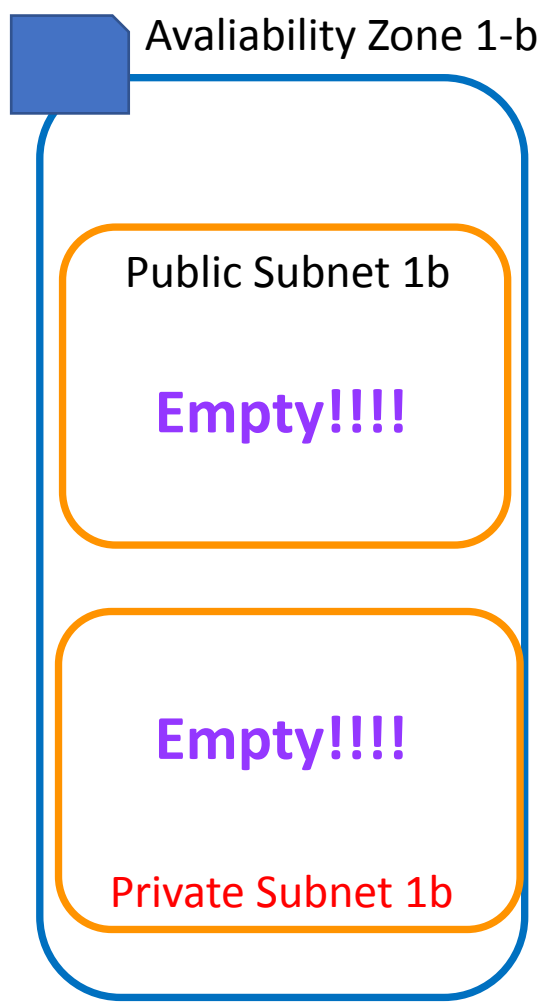
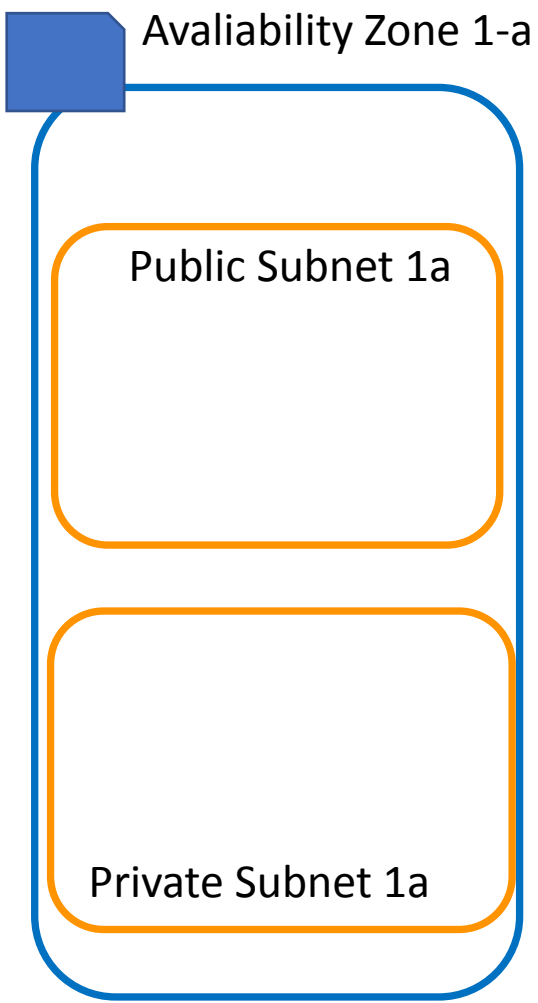


VPC

2- Where we are



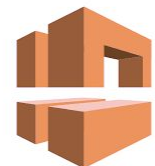
Internet Gateway





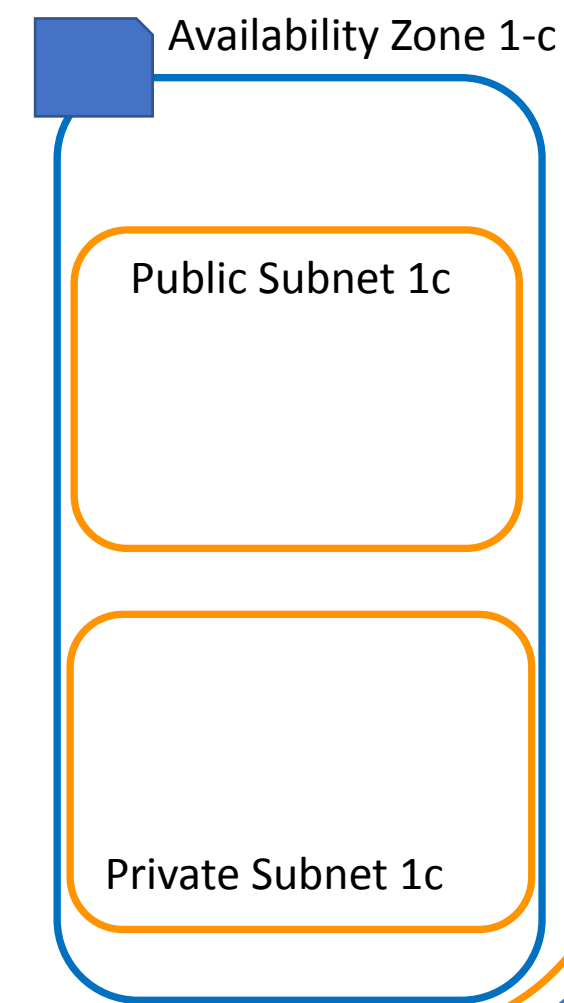
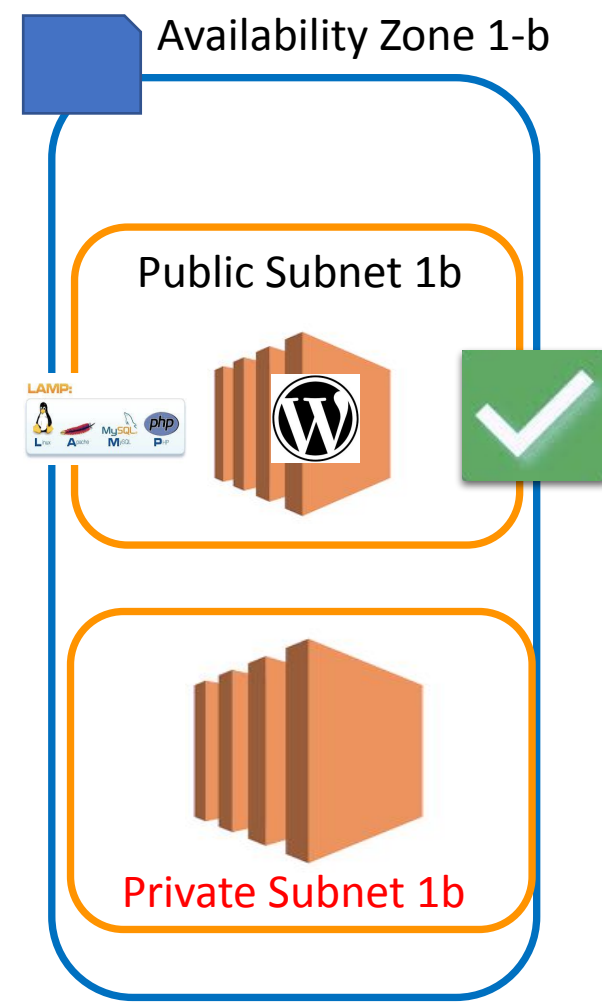
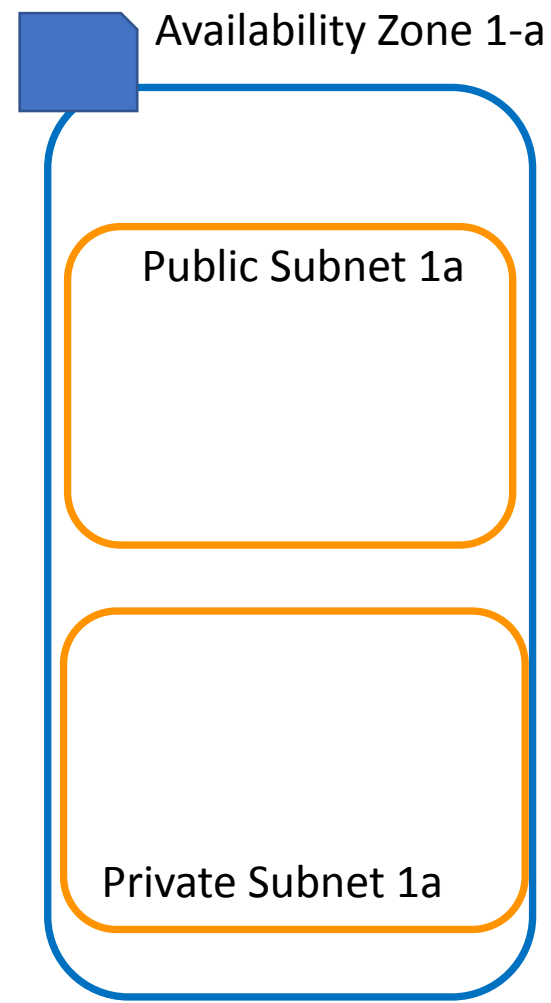
Cloud

Region



VPC

3- Wordpress Instance is ready what about DB



Security Group Best Practice

Bastion Host

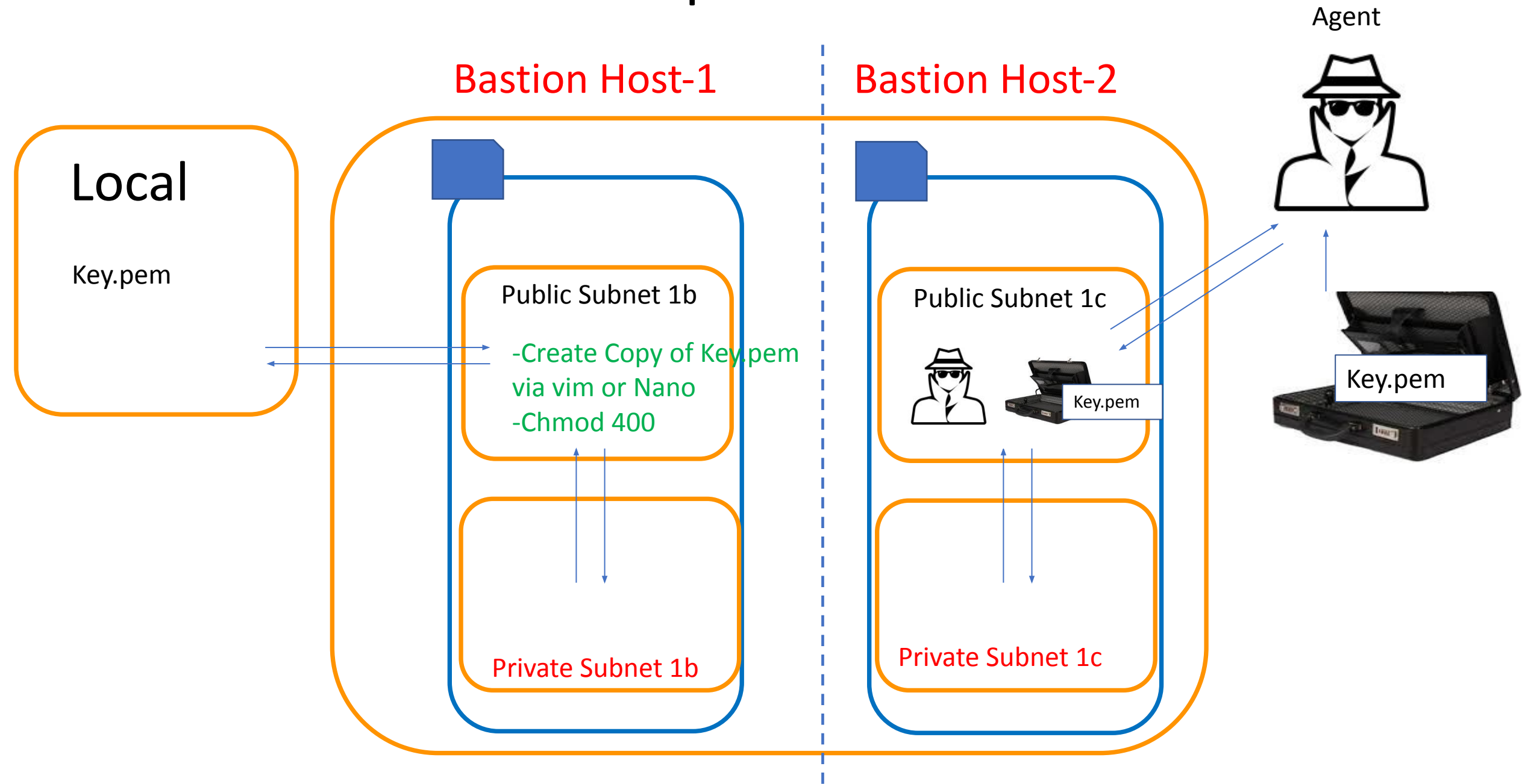
Inbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
All traffic ▼	All	All	Custom ▼		Delete



- 1-Sec. group of Bastion Host –Best practice
- 2-CIDR Block of “Public Subnet”
- 3-IP of Bastion Host Instance

.pem Issue





Cloud

Internet Gateway

Region



VPC

3- You are here now

Availability Zone 1-a

Availability Zone 1-b

Availability Zone 1-c

Public Subnet 1a

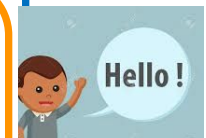
Public Subnet 1b

Public Subnet 1c

Private Subnet 1a

Private Subnet 1b

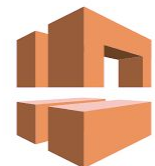
Private Subnet 1c





Cloud

Region



VPC



Internet Gateway

3- Try to install mariaDB



Public Subnet 1a

Private Subnet 1a

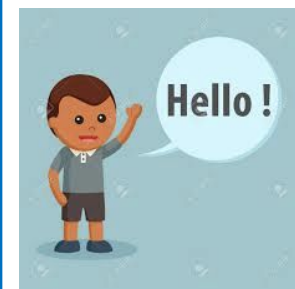


Public Subnet 1b

Private Subnet 1b

1
Bastion Host

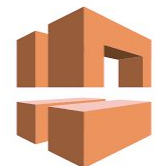
2
Install mariadb





Cloud

Region

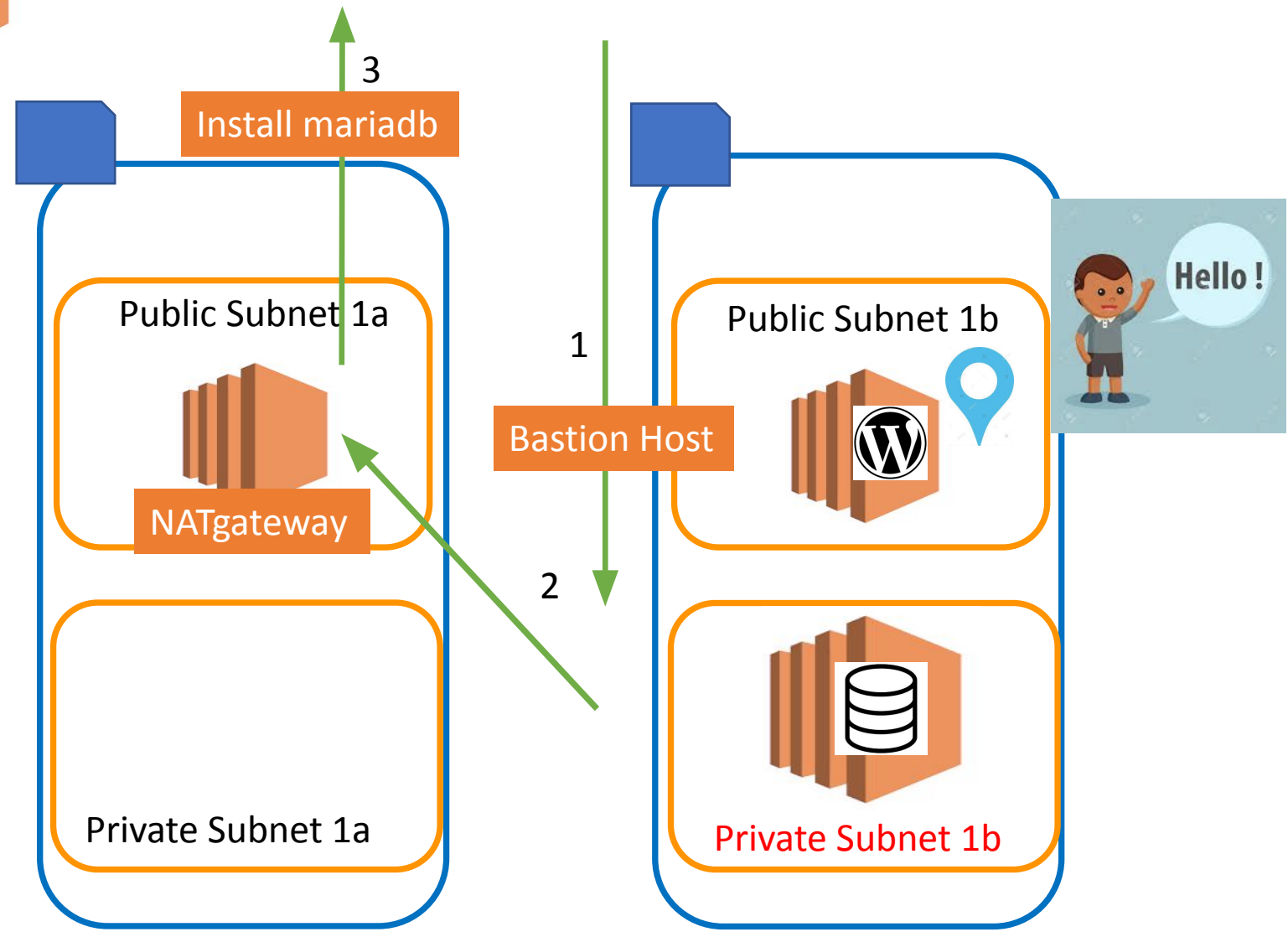


VPC



Internet Gateway

4- Try Nat instance



NAT INSTANCE

[Route Tables](#) > Edit routes

Edit routes

1- Route table Issue

Destination	Target	Status	Propagated	
10.0.0.0/16	local	active	No	
0.0.0.0/0	i-05aeca8f8ef883dec		No	✕

Add route



- Nat instance

2- Change Source/ Destination Check

- Disable



Cloud



Region

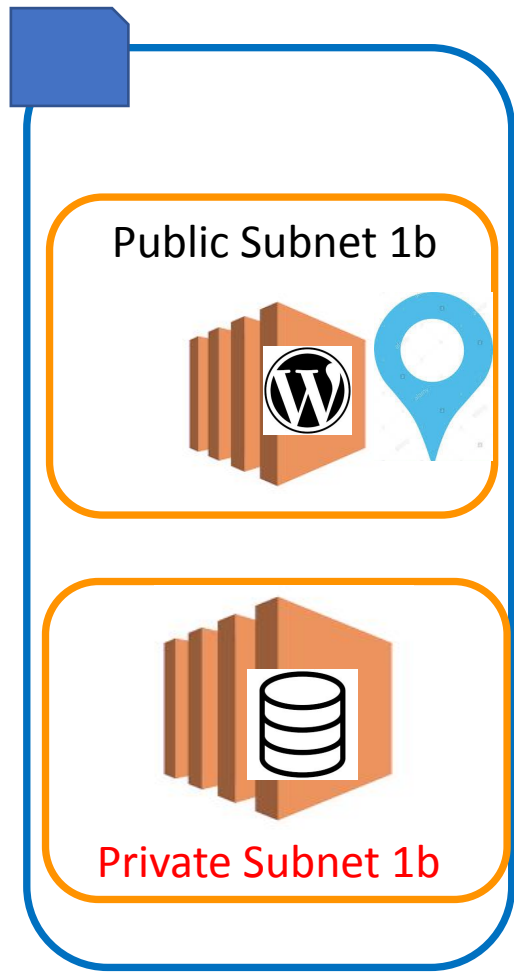
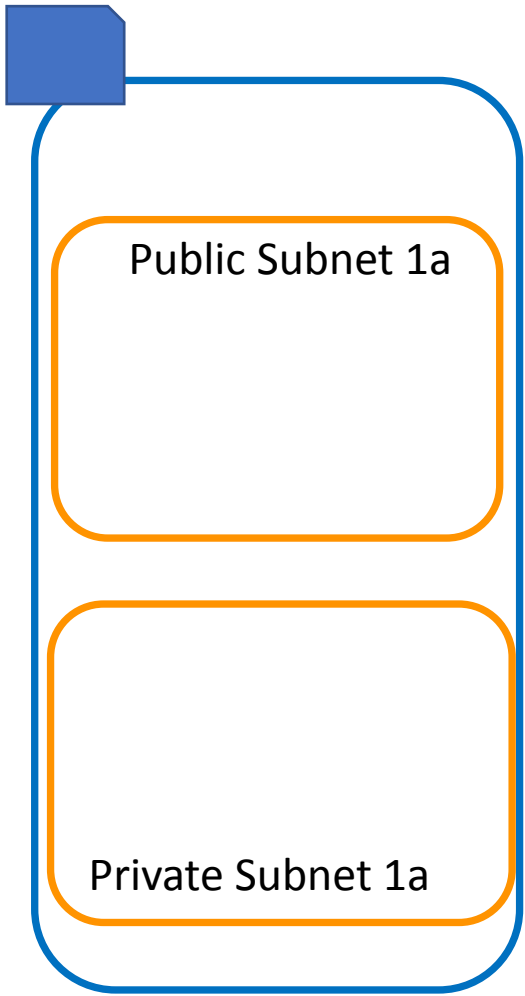


VPC



Internet Gateway

5- Associate DATABASE

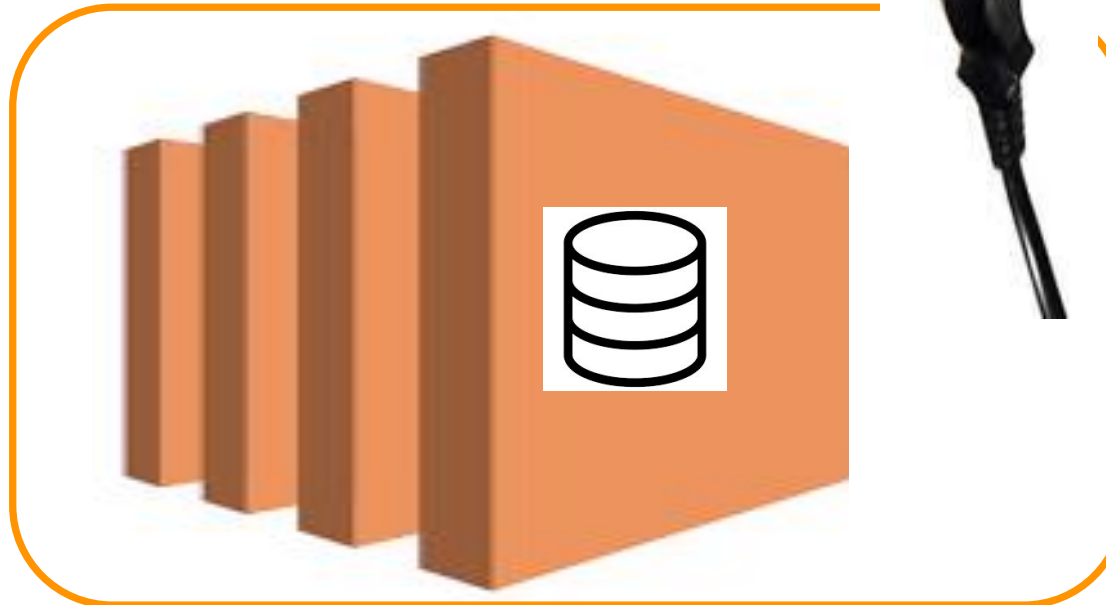


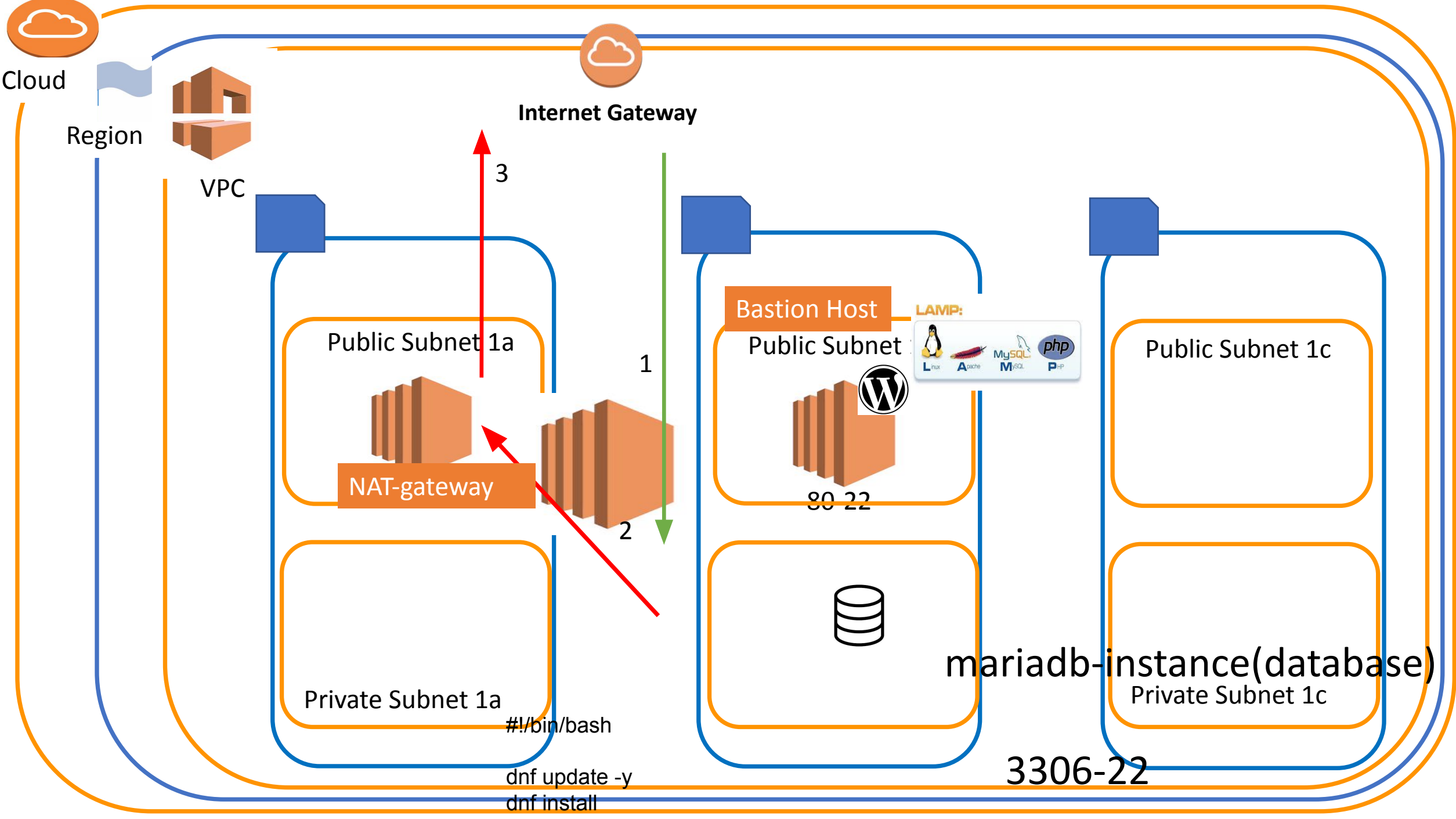
Associate DATABASE

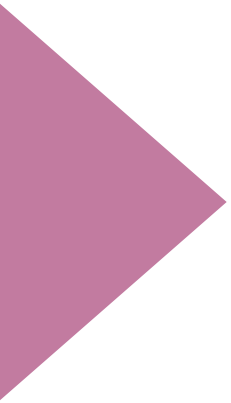
Public Subnet 1b



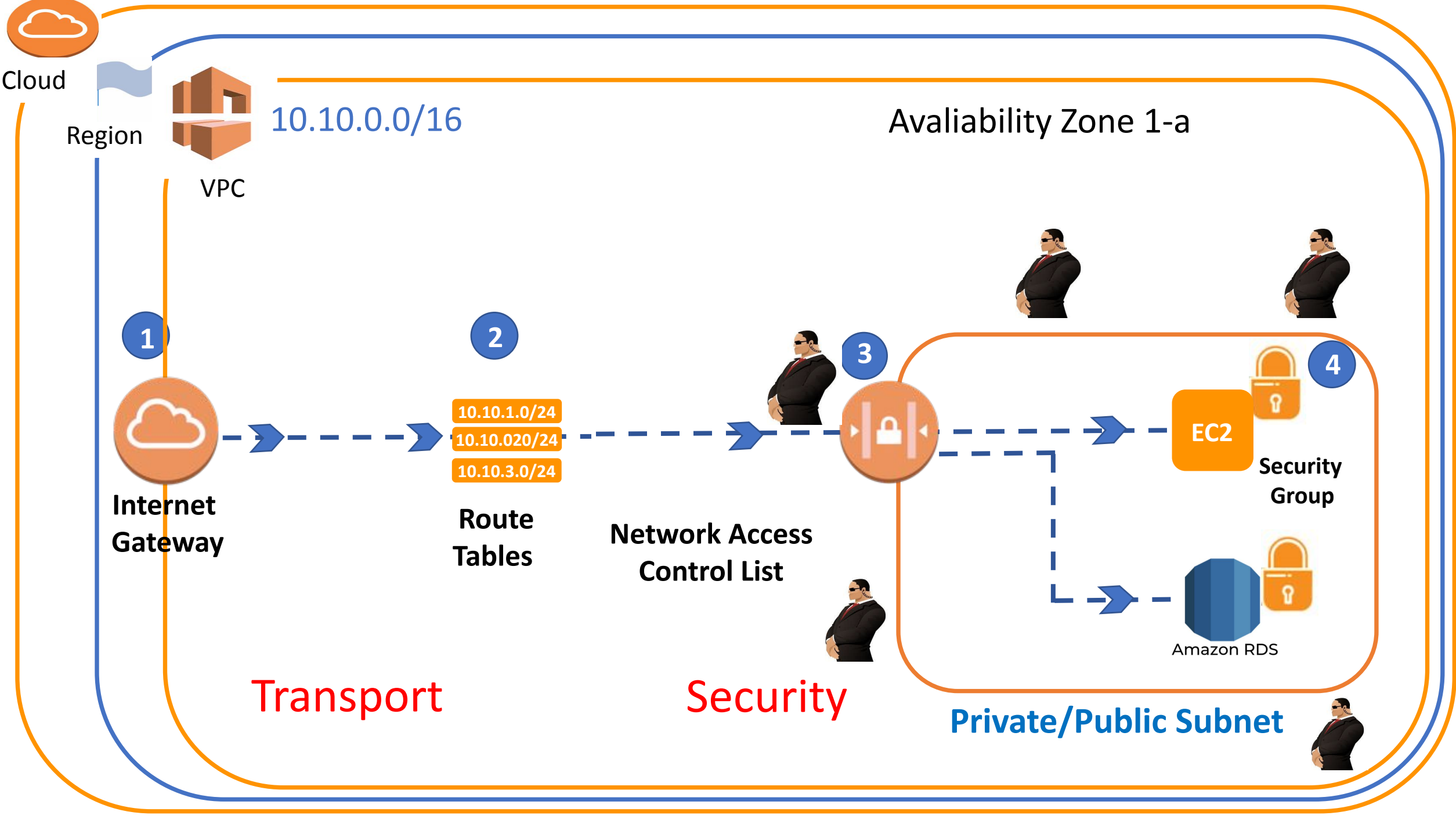
Private Subnet 1b

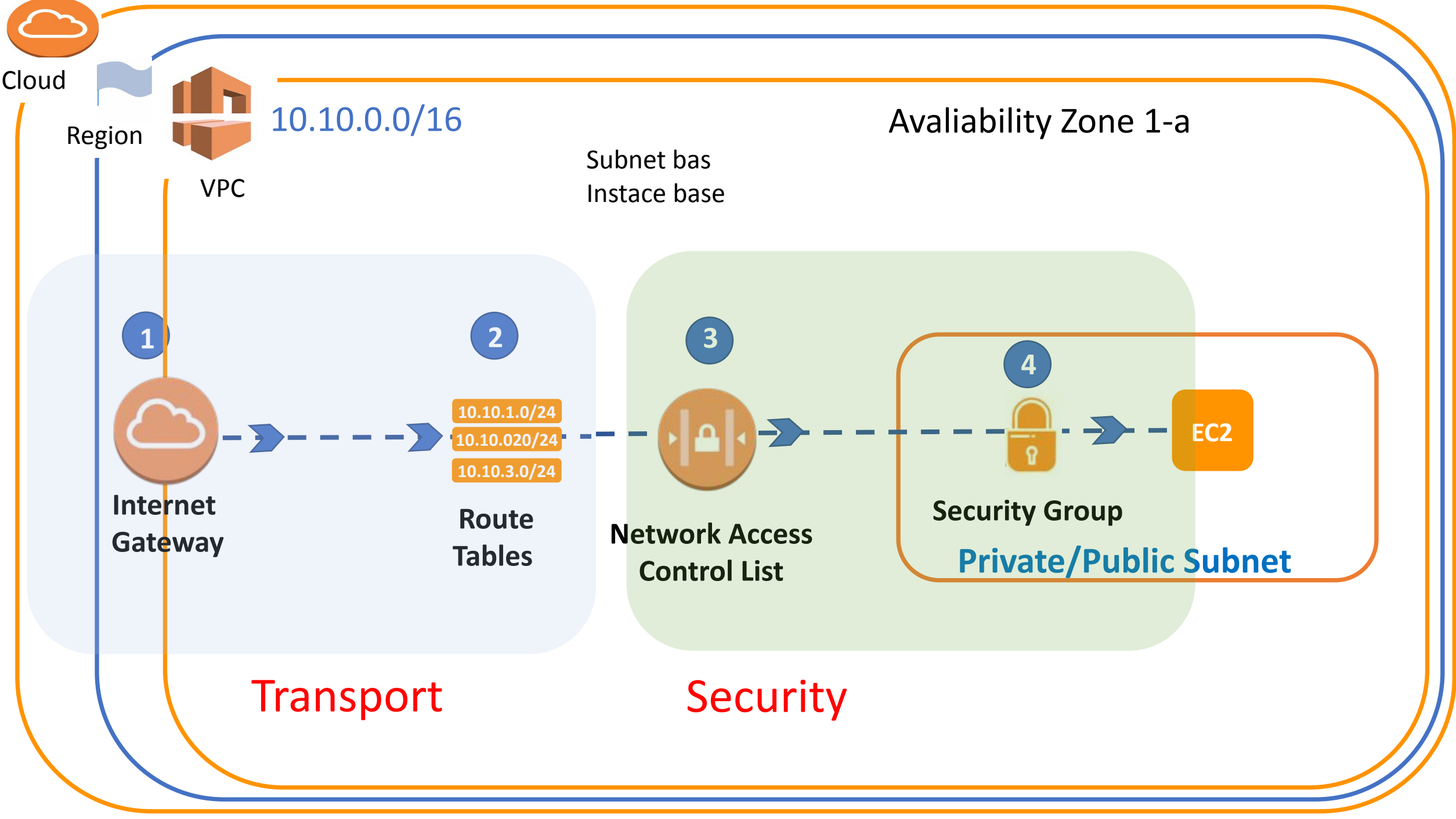






NACL (NETWORK ACCESS LISTS)

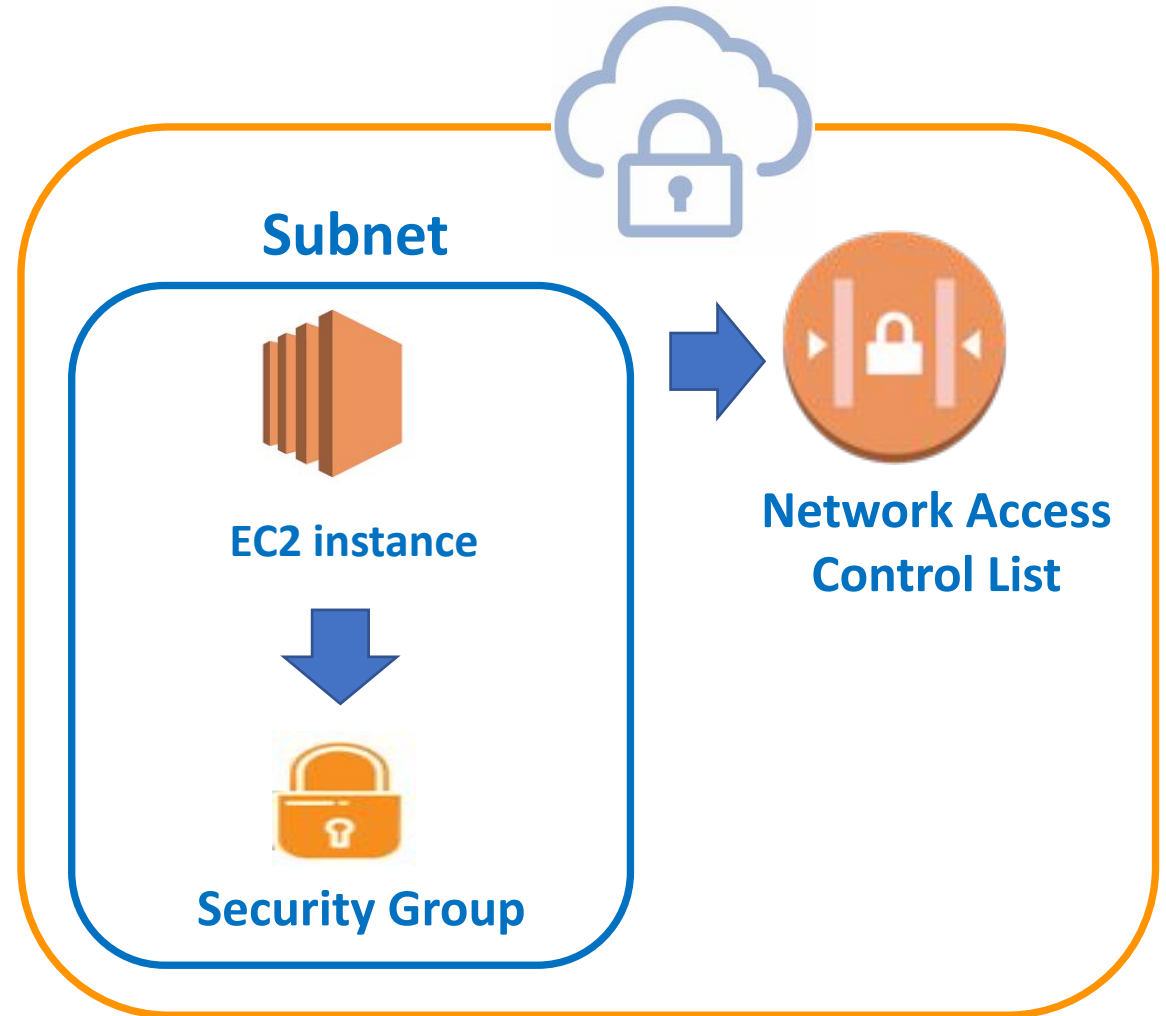




NACL (NETWORK ACCESS LISTS)

Subnet obeys the NACL rules

Resources obeys NACL and Sec. Group



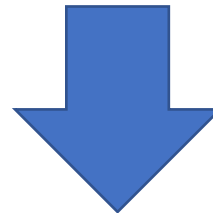
(Statefull) Security Group inbound

Type	Protocol	Port Range	Source
HTTP	TCP(6)	80	1.2.3.4/32
SSH-22	TCP(6)	22	0.0.0.0/0
All ICMP-IPv4	ICMP(1)	ALL	0.0.0.0/0
HTTPS	TCP(6)	443	7.8.9.10/32

ALLOW Only

Network ACL inbound (Stateless)

Rule	Type	Protocol	Port Range	Source	Allow/Deny
100	HTTP	TCP(6)	80	7.8.9.10/32	ALLOW
200	SSH-22	TCP(6)	22	0.0.0.0/0	ALLOW
300	All ICMP-IPv4	ICMP(1)	ALL	0.0.0.0/0	ALLOW
400	HTTPS	TCP(6)	443	7.8.9.10/32	DENY
*	ALL Traffic	ALL	ALL	0.0.0.0/0	DENY



(Stateless) Network ACL outbound

Rule	Type	Protocol	Port Range	Destination	Allow/Deny
100	HTTP	TCP(6)	80	7.8.9.10/32	ALLOW
200	Custom TCP	TCP(6)	32768 -65535	0.0.0.0/0	ALLOW
300	All ICMP-IPv4	ICMP(1)	ALL	0.0.0.0/0	ALLOW
400	HTTPS	TCP(6)	443	7.8.9.10/32	DENY
*	ALL Traffic	ALL	ALL	0.0.0.0/0	DENY



PC IP: 7.8.9.10/32

Connection Request

No	Type-Port
1	SSH-22
2	HTTP-80
3	All ICMP-IPv4 -All
4	HTTPS-443
5	Msql/Auro. 3306



Security Group inbound

Type	Protocol	Port Range	Source
HTTP	TCP(6)	80	1.2.3.4/32
SSH-22	TCP(6)	22	0.0.0.0/0
All ICMP-IPv4	ICMP(1)	ALL	0.0.0.0/0
HTTPS	TCP(6)	443	7.8.9.10/32



Subnet

Network ACL in/outbound

Rule	Type	Protocol	Port Range	Source/ Destination	Allow/ Deny
100	HTTP	TCP(6)	80	7.8.9.10/32	ALLOW
200	SSH-22	TCP(6)	22	0.0.0.0/0	ALLOW
300	All ICMP-IPv4	ICMP(1)	ALL	0.0.0.0/0	ALLOW
400	HTTPS	TCP(6)	443	7.8.9.10/32	DENY
*	ALL Traffic	ALL	ALL	0.0.0.0/0	DENY



User IP: 7.8.9.10/32

Connection
Request

No	Type-Port
1	SSH-22
2	HTTP-80
3	All ICMP-IPv4 -All
4	HTTPS-443
5	Msql/Auro. 3306



Security Group inbound

Type	Protocol	Port Range	Source
HTTP	TCP(6)	80	1.2.3.4/32
SSH-22	TCP(6)	22	0.0.0.0/0
All ICMP-IPv4	ICMP(1)	ALL	0.0.0.0/0
HTTPS	TCP(6)	443	7.8.9.10/32



Network ACL in/outbound

Rule	Type	Protocol	Port Range	Source/ Destination	Allow/ Deny
100	HTTP	TCP(6)	80	7.8.9.10/32	ALLOW
200	SSH-22	TCP(6)	22	0.0.0.0/0	ALLOW
300	All ICMP-IPv4	ICMP(1)	ALL	0.0.0.0/0	ALLOW
400	HTTPS	TCP(6)	443	7.8.9.10/32	DENY
*	ALL Traffic	ALL	ALL	0.0.0.0/0	DENY



User IP: 7.8.9.10/32

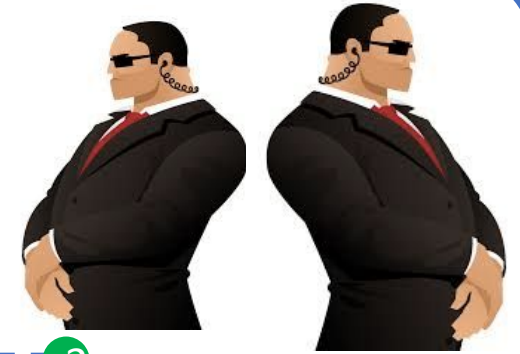
Connection
Request

No	Type-Port
1	SSH-22
2	HTTP-80
3	All ICMP-IPv4 -All
4	HTTPS-443
5	Msql/Auro. 3306



Security Group inbound

Type	Protocol	Port Range	Source
HTTP	TCP(6)	80	1.2.3.4/32
SSH-22	TCP(6)	22	0.0.0.0/0
All ICMP-IPv4	ICMP(1)	ALL	0.0.0.0/0
HTTPS	TCP(6)	443	7.8.9.10/32



Network ACL in/outbound

Rule	Type	Protocol	Port Range	Source/ Destination	Allow/ Deny
100	HTTP	TCP(6)	80	7.8.9.10/32	ALLOW
200	SSH-22	TCP(6)	22	0.0.0.0/0	ALLOW
300	All ICMP-IPv4	ICMP(1)	ALL	0.0.0.0/0	ALLOW
400	HTTPS	TCP(6)	443	7.8.9.10/32	DENY
*	ALL Traffic	ALL	ALL	0.0.0.0/0	DENY



User IP: 7.8.9.10/32

Connection
Request

No	Type-Port
1	SSH-22
2	HTTP-80
3	All ICMP-IPv4 -All
4	HTTPS-443
5	Msql/Auro. 3306



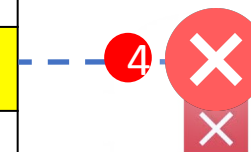
Security Group inbound

Type	Protocol	Port Range	Source
HTTP	TCP(6)	80	1.2.3.4/32
SSH-22	TCP(6)	22	0.0.0.0/0
All ICMP-IPv4	ICMP(1)	ALL	0.0.0.0/0
HTTPS	TCP(6)	443	7.8.9.10/32



Network ACL in/outbound

Rule	Type	Protocol	Port Range	Source/ Destination	Allow/ Deny
100	HTTP	TCP(6)	80	7.8.9.10/32	ALLOW
200	SSH-22	TCP(6)	22	0.0.0.0/0	ALLOW
300	All ICMP-IPv4	ICMP(1)	ALL	0.0.0.0/0	ALLOW
400	HTTPS	TCP(6)	443	7.8.9.10/32	DENY
*	ALL Traffic	ALL	ALL	0.0.0.0/0	DENY





User IP: 7.8.9.10/32

Connection
Request

No	Type-Port
1	SSH-22
2	HTTP-80
3	All ICMP-IPv4 -All
4	HTTPS-443
5	Msql/Auro. 3306



Security Group inbound

Type	Protocol	Port Range	Source
HTTP	TCP(6)	80	1.2.3.4/32
SSH-22	TCP(6)	22	0.0.0.0/0
All ICMP-IPv4	ICMP(1)	ALL	0.0.0.0/0
HTTPS	TCP(6)	443	7.8.9.10/32



Network ACL in/outbound

Rule	Type	Protocol	Port Range	Source/ Destination	Allow/ Deny
100	HTTP	TCP(6)	80	7.8.9.10/32	ALLOW
200	SSH-22	TCP(6)	22	0.0.0.0/0	ALLOW
300	All ICMP-IPv4	ICMP(1)	ALL	0.0.0.0/0	ALLOW
400	HTTPS	TCP(6)	443	7.8.9.10/32	DENY
*	ALL Traffic	ALL	ALL	0.0.0.0/0	DENY

5

5

5

5

5

5



EPHEMERAL PORT

NACLs are stateless. This means that you are required to have a rule for inbound AND outbound traffic. So, if you want to allow your EC2 instance to serve HTTP traffic, you will need to allow port 80 inbound and ports 1024 – 65535 outbound. But where 1024 – 65535 came from.

The ports 1024 – 65535 are called the “ephemeral ports”.

These ports are randomly selected to allow return traffic for a request. So, if a request comes to the server on port 80, the request also specifies a random port between 1024 – 65535 for the return traffic.

NACL TABLES

Let's get our hands dirty!

- NACL Tables

THANKS!

Any questions?

